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(54) **LARGE LANGUAGE MODELS UTILIZING ELEMENT-WISE OPERATION-FUSION**

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(57) **ABSTRACT**

Techniques for using a quantized and/or fused model are described. In some examples, a service is to receive a request to use a trained model, the request including input data; apply a trained model to the input data, the application of the trained model includes fusing one or more matrix multiplication operations with element-wise operations; and output a result from the trained model.

20 Claims, 14 Drawing Sheets

